GlucoTwin

Predictive Metabolic Digital Twin for Hypoglycemia Risk

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CRP Digital Twin Challenge 2026 · Happiest Health

Problem Statement

- Hypoglycemia (≤ 70 mg/dL) is the most critical acute hazard for Type 1 Diabetes patients
- Risk is highest during sleep — when symptom perception is absent
- Current CGM threshold alerts: 29.21 raw false alarms/day causing severe alert fatigue
- Static alarms ignore physiological context: IOB, COB, circadian vulnerability
- No counterfactual simulation: cannot ask 'what if the patient takes 15g carbs?'
- No continuous patient memory: every moment evaluated in isolation

Healthcare Use Case

- Target Users: Type 1 Diabetes patients on CSII/MDI insulin therapy with CGM sensors
- Clinical Setting: Nocturnal hypoglycemia prevention and early-warning decision support
- Clinician Decision Support: Calibrated 60-min risk alerts before threshold crossing
- Patient Empowerment: Interactive What-If simulator for rescue carb intervention planning
- Hospital/Clinic Use: Multi-patient dashboard with per-patient personalized twin states
- Regulatory Scope: Research prototype — clinician decision support only (not FDA-cleared)

Technical Stack

ML & Backend

- Python 3.10/3.11
- LightGBM + RandomForest ensemble
- SHAP (explainability)
- Scikit-learn (isotonic calibration)
- NumPy / Pandas / SciPy
- Joblib (model serialization)
- pytest (38 tests, CI verified)

Frontend & Infrastructure

- Streamlit (clinical dashboard)
- Plotly (interactive charts)
- GitHub Actions CI/CD
- Python 3.10 + 3.11 CI matrix
- edge-tts / moviepy (demo video)
- python-pptx / reportlab (docs)
- Dataset: HUPA-UCM (Mendeley DOI)

AI/ML Model & Framework Details

- Primary Model: LightGBM 60-min hypoglycemia classifier (50 features)
- Validation: Leave-One-Patient-Out (LOPO) on 24 patients — ROC-AUC 0.970 ± 0.027
- Calibration: Isotonic regression — Brier score 0.069 → 0.039 (−43.5%), ECE −94.5%
- Explainability: SHAP TreeExplainer — real-time top-K risk factor attribution
- Trajectory: Damped-velocity ODE ($k=0.015\ min^{-1}$) — TTT MAE 32.55 min over 11,471 pairs
- Alert FSM: Hysteresis state machine — 4.39 episodes/day vs 29.21 raw point alarms
- OOD Detection: Mahalanobis distance + CGM continuity → twin_confidence 0.0–1.0

5 Core Digital Twin Differentiators

- ① Continuous Patient Memory — persistent episode log with strict backward temporal guards
- ② Trajectory Modeling with Validated Uncertainty — damped ODE + empirical MAE tiers
- ③ Counterfactual What-If Simulation — in-silico rescue carb & bolus scenario engine
- ④ Stateful Alert Engine with Hysteresis — 4.39 episodes/day (vs 29.21 raw false alarms)
- ⑤ Model Self-Awareness & OOD Guardrails — TwinState v6 exposes full empirical metrics

Authoritative Empirical Results

- LOPO ROC-AUC: 0.970 ± 0.027 | PR-AUC: 0.858 ± 0.099 (24 patients, patient-independent)
- LOPO Sensitivity: 0.958 ± 0.041 | Specificity: 0.821 ± 0.088 | F2: 0.754 ± 0.065
- TTT MAE: 32.55 min | Median AE: 17.60 min | Within 15 min: 46.0% (11,471 pairs)
- Brier Score: 0.0690 (raw) → 0.0390 (calibrated) | ECE: 0.1006 → 0.0055
- Full-Cohort Ablation AUC: 0.9468 | PR-AUC: 0.8132 | Sensitivity: 0.8628
- Episode Detection: 300/300 (100%) | Mean Lead Time: 47.4 min
- Alert Episodes: 4.39/day (vs naive 29.21 raw point-level alarms/day)

Streamlit Clinical Dashboard (app.py)

- Live Inference Mode: real-time risk scoring from production LightGBM + IsotonicCalibrator
- Patient Navigator: Sidebar dropdown for all 25 HUPA-UCM cohort patients
- Current State Panel: glucose, 15-min velocity, risk score, alert status, trend
- What-If Simulator: carb intervention sliders (0–80g), bolus, exercise, 30–180 min horizon
- SHAP Explanations: top risk drivers computed live per inference call
- OOD Confidence Panel: twin_confidence score + Mahalanobis distance visualization
- Demo Fallback Mode: verified static payload if model artifacts are absent

System Architecture Overview

- Data Layer: HUPA-UCM CGM + Bolus + Meal + Activity data ingestion
- Processing Layer: Leakage-safe preprocessing → 50-feature engineering pipeline
- Twin Core: LightGBM + Isotonic Calibrator → Alert FSM → Patient Memory → OOD
- Simulation Layer: MetabolicSimulator v2 (Bergman 2-compartment ODE)
- API Layer: GlucoTwinAPI → TwinState v6 JSON (full snapshot, SHAP, What-If)
- Presentation Layer: Streamlit + Plotly live dashboard (app.py)
- Full architecture diagram: docs/submission/GlucoTwin_Architecture_Diagram.pdf

Limitations & Ethical Considerations

- RESEARCH PROTOTYPE — NOT A MEDICAL DEVICE (not FDA/CE cleared)
- Observational cohort: 25 Spanish adults with T1D — generalizability unknown
- Sensor-specific: FreeStyle Libre 2 CGM; other devices may differ in factory calibration
- No Closed-Loop Control: open-loop decision support only — not integrated into AID systems
- What-If simulations are mathematical projections, NOT clinical prescriptions
- All predictions are statistical probabilities — never replace physician judgment

Team GlucoTwin

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github.com/Ronak1167/glucotwin | CI: ✅ 38/38 tests passing